

Feiyu Du

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Education

The University of Texas at Dallas
Ph.D. in Computer Science, Advisor: Dingzhu Du

Started on 08/2026
Richardson, Texas, US

Shaanxi University of Science & Technology
M.S. in Software Engineering

09/2021 - 06/2024
Shaanxi, CHN

Shaanxi University of Science & Technology
B.S. in Information and Computational Science

09/2017 - 06/2021
Shaanxi, CHN

Experience

Research Assistant, The University of Texas at Dallas
Topic: Multimodal Learning and Egocentric Video Understanding

06/2025 - Present
Dallas, US

- Developed multimodal learning systems for egocentric video understanding by integrating visual, audio, speech, and temporal information for social interaction reasoning.
- Developed online audio-visual learning models in **PyTorch** using visual and audio encoders, temporal sequence modeling, and multimodal feature fusion for real-time social interaction understanding.
- Designed a large-scale egocentric social intelligence benchmark for evaluating multimodal foundation models across speech activity, speaker grounding, addressee reasoning, referent understanding, wearer participation, and interaction dynamics.
- Built reproducible pipelines for multimodal QA construction, dataset validation, benchmark auditing, model inference, and quantitative evaluation.
- Conducted robustness and generalization analysis under missing modalities, visual detection noise, and imperfect real-world observations.
- Led research on online Talk-to-Me detection in egocentric videos, resulting in a paper accepted to **ACM Multimedia 2026**.

Research Assistant, University of Electronic Science and Technology of China
Topic: Quantum Machine Learning and Domain Adaptation

07/2024 - 03/2025
Chengdu, CHN

- Designed and implemented quantum-enhanced machine learning algorithms for domain adaptation and transfer learning under heterogeneous and noisy quantum environments.
- Investigated representation alignment, distribution shift, feature interaction, and model generalization in quantum machine learning systems.
- Implemented and evaluated learning algorithms across simulated and hardware-oriented quantum settings, leading to multiple research works in quantum machine learning.

Research Papers

[1] Calibration Is Not Recoverability: A-Priori Certificates for Amortized Simulation-Based Inference

Xi He, Feiyu Du

AAAI 2027, Under Review

[2] Estimating Feature Interactions Across Input Distributions in Quantum Machine Learning

Xi He, Feiyu Du

AAAI 2027, Under Review

[3] Talking to Me or Someone Else? Rethinking Talk-to-Me Detection in Egocentric Videos

Feiyu Du, Xi He, Jia Li, Yapeng Tian, Weili Wu

ACM MM 2026, Accepted

[4] Path-Length-Aware Gradual Domain Adaptation via Edge-Budgeted Local Drift Control

Xi He, Feiyu Du, Xiaohan Yu, Weili Wu

NeurIPS 2026, Under Review

[5] Representation Reuse as an Observation Channel: Common-Measurement Complexity in Repeated Quantum Inverse Learning

Xi He, Feiyu Du, Yapeng Tian, Weili Wu

NeurIPS 2026, Under Review

[6] Do Joint Audio-Video Generation Models Understand Physics?

Zijun Cui, Xiulong Liu, Hao Fang, Mingwei Xu, Jiaqeng Liu, Zexin Xu, Weiguo Pian, Shijian Deng, Feiyu Du, Chenming Ge, Yapeng Tian

NeurIPS 2026 E&D, Under Review

[7] Fish Target Recognition Method Based on Joint Deep Statistical Feature Alignment

Haiyan Wang, Feiyu Du, Haiyang Yao, Xiao Chen

JSUST, 2024

[8] Underwater Image Enhancement Algorithm Based on Domain Adaptive Method

Feiyu Du, Haiyan Wang, Haiyang Yao, Xiao Chen

Computer and Modernization, 2023

[9] Quantum Joint Distribution Adaptation on the Universal Quantum Computer

Feiyu Du, Xi He

IEEE PRAI 2023

[10] Quantum Kernel Subspace Alignment for Unsupervised Domain Adaptation

Xi He, Feiyu Du

ACML 2023

[11] Quantum Classifiers for Domain Adaptation

Xi He, Feiyu Du, Mingyuan Xue, Xiaogang Du, Tao Lei, A. K. Nandi

Quantum Information Processing, 2023

Technical Skills

Languages: Python, C/C++, SQL

Technologies: PyTorch, scikit-learn, Huggingface, PyTorch-Lightning, OpenCV, Git, Linux, CUDA, LaTeX

Research & Data: Benchmark construction, data pipelines, dataset validation, model evaluation, relational databases, transfer learning, domain adaptation